
Thermal-Based Human Recognition via Spiking Neural Networks for Low-Power Embedded Systems

Research Progress Presentation | August 6 2026

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| Presentation Roadmap

WHY

**Physical AI
Motivation
Objective**

HOW

**System & data
SNN encoding
CNN / SNN
models**

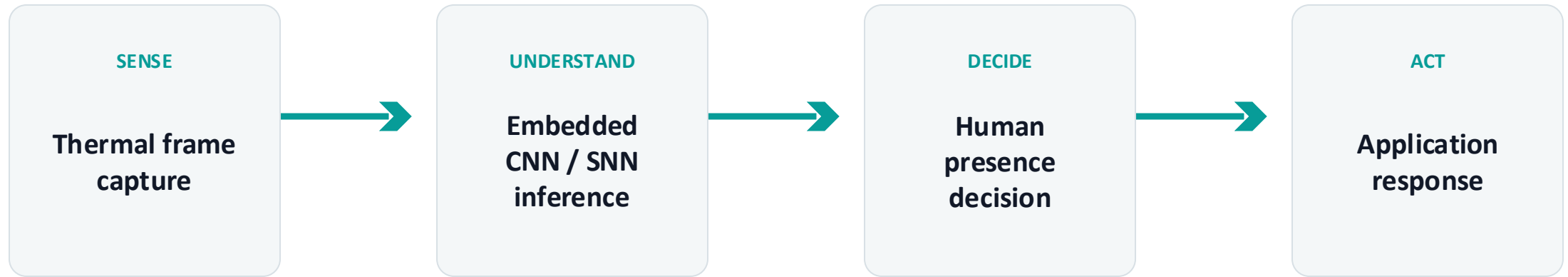
EVIDENCE

**Static test
Dynamic CNN
Pi deployment**

NEXT

**Limitations
Power & energy
Thesis
completion**

| Physical AI and Human Perception



Privacy-aware detection ▪ Embedded device

| Why Thermal Sensing at the Edge?

VISIBLE CAMERA

**Identity-visible,
lighting-dependent
images**

32 × 24 THERMAL

**Privacy-preserving
32 × 24 heat maps**

EDGE INFERENCE

**Low-latency
processing
without cloud access**

| Research Objective

CNN vs SNN on Raspberry Pi 5

SAME INPUT

32 × 24 thermal frames

COMPARE QUALITY

Accuracy, precision, recall, F1

SAME CONDITIONS

Dataset and preprocessing

COMPARE COST

Latency, power, energy

| Related Work and Research Position

01 Thermal sensing for presence, localization, and crowd counting

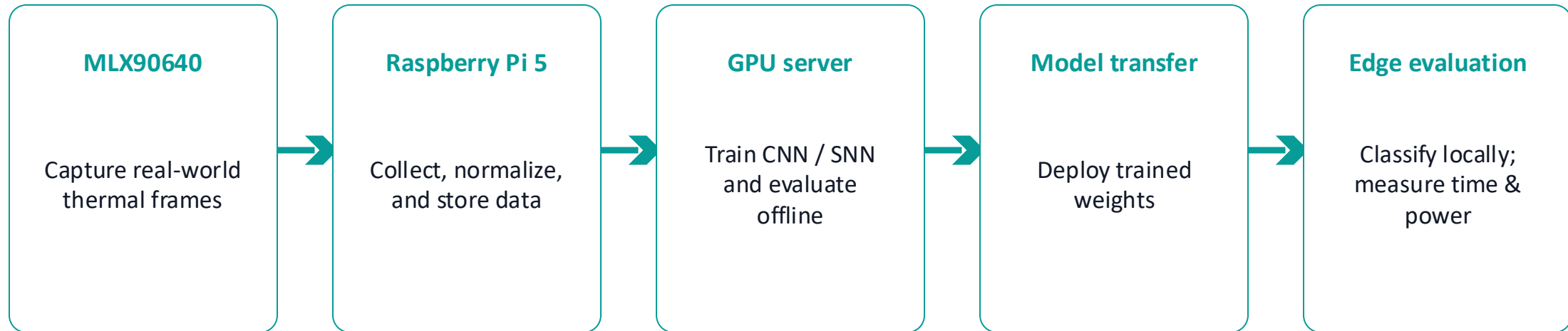
02 Embedded CNNs for edge-based human detection

03 Trainable SNNs using surrogate gradients

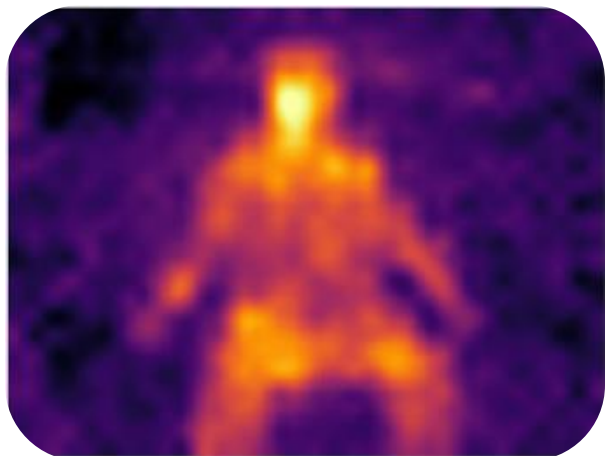
Gap addressed

Same data ▪ Same preprocessing ▪ Same Raspberry Pi

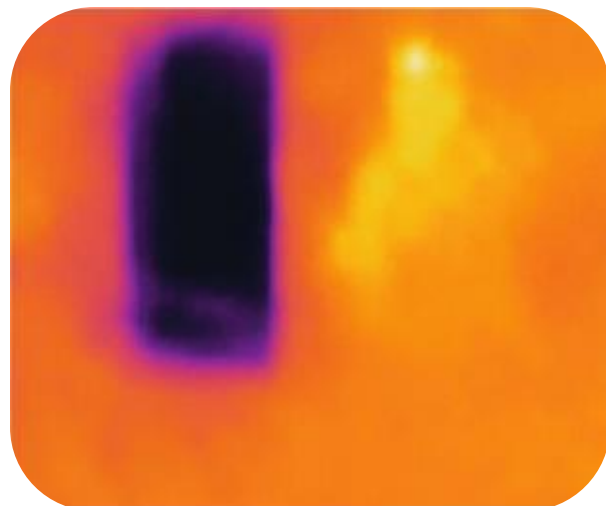
| System Workflow



Dataset: 1,800 Real-World Thermal Frames



Human | 800



No Human | 1,000

70%

Training

15%

Validation

15%

Test

Random seed: 42
No stratified split

RELATIVE TEMPERATURE

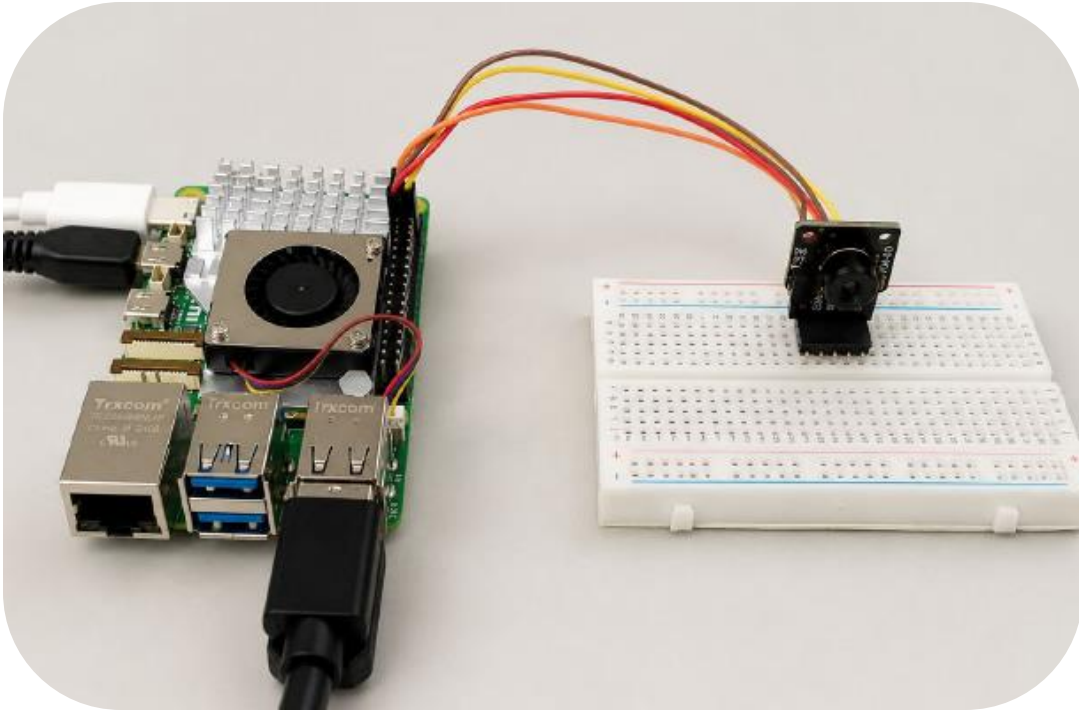


Lower

Higher

Per-frame normalization
No absolute-temperature scale

Hardware Selection



MLX90640

32 × 24 ▪ Wide FoV ▪ Privacy

Raspberry Pi 5

CNN / SNN ▪ Edge inference

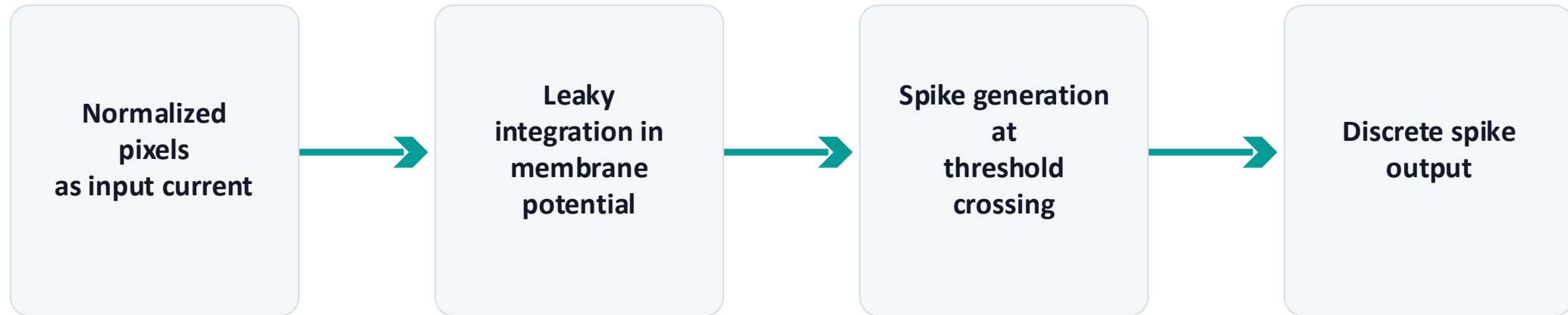
GPU server

Training ▪ Offline evaluation

Power meter

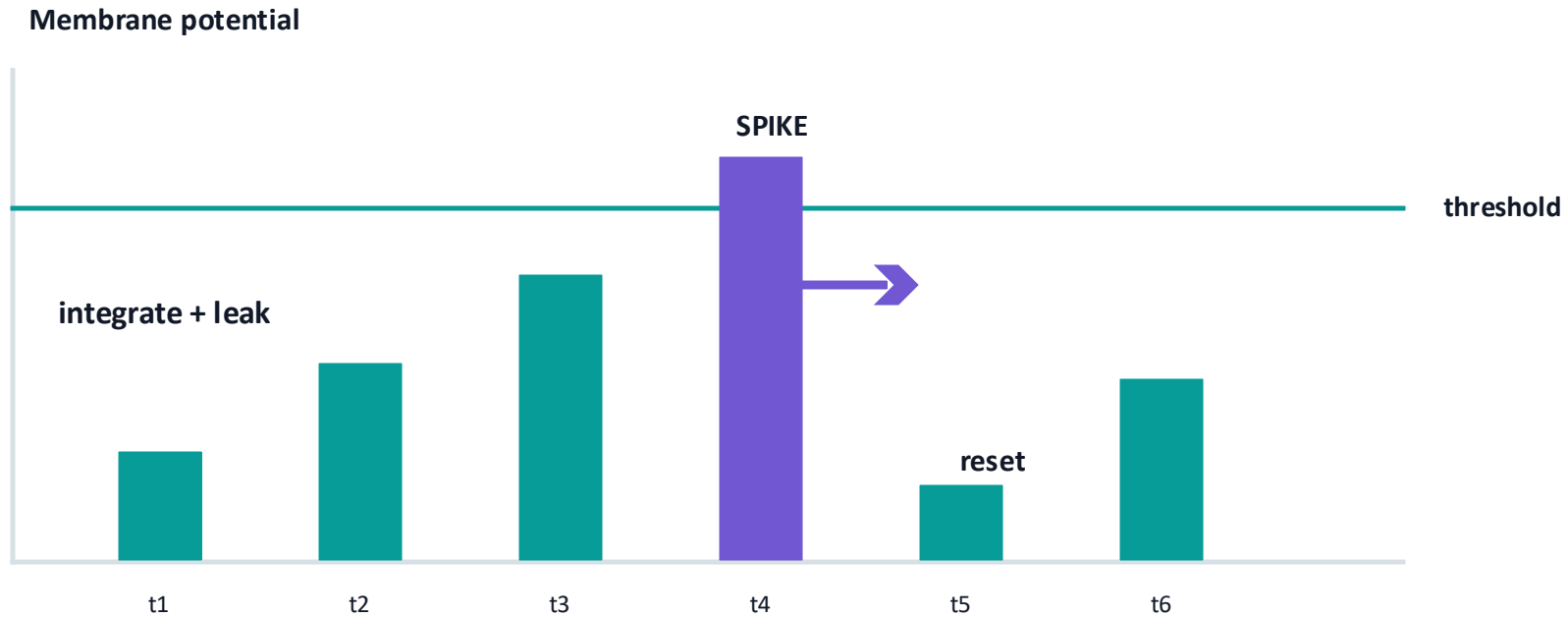
Setup: indoor lab | sensor height ≈ 60 cm | range 0–5 m | FoV $110^\circ \times 75^\circ$

| Spiking Neural Network



Potential for sparse computation

Leaky Integrate-and-Fire Neuron



- Integrate input
- Leak over time
- Fire at threshold
- Reset after spike

25 internal timesteps

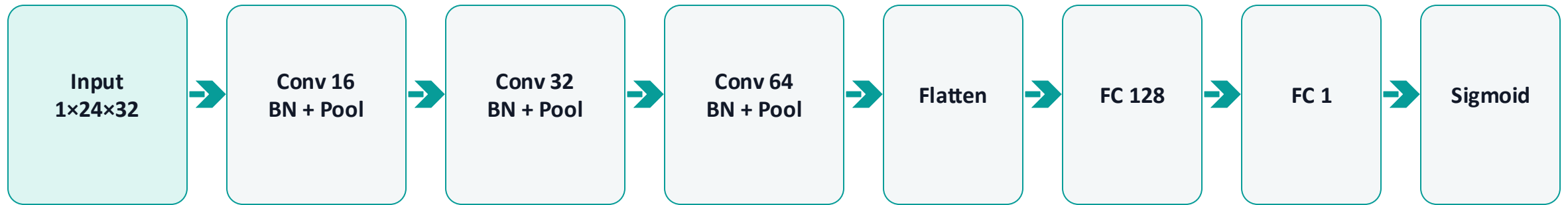
| Direct Encoding: One Frame, 25 Timesteps



DIRECT ENCODING

LIF layers generate spikes

CNN Architecture



Single pass ▪ 3 convolution stages ▪ 2 fully connected layers

| SNN Architecture



| Static Test Results

0.9912

CNN F1

1.0000

SNN F1

270

Test frames

Frame-level split with similar neighboring frames

Generalization to new subjects and rooms is not established

| Dynamic CNN Results

253

TP

47

FN

2

FP

298

TN

0.9183

Accuracy

0.9117

F1

600 separately collected frames (300 Human / 300 No Human) | Threshold: 0.50

Only 2 false positives, but 47 / 300 human frames were missed

Threshold Analysis

Threshold 0.50

47

False negatives

2

False positives



Threshold 0.12

10

False negatives

13

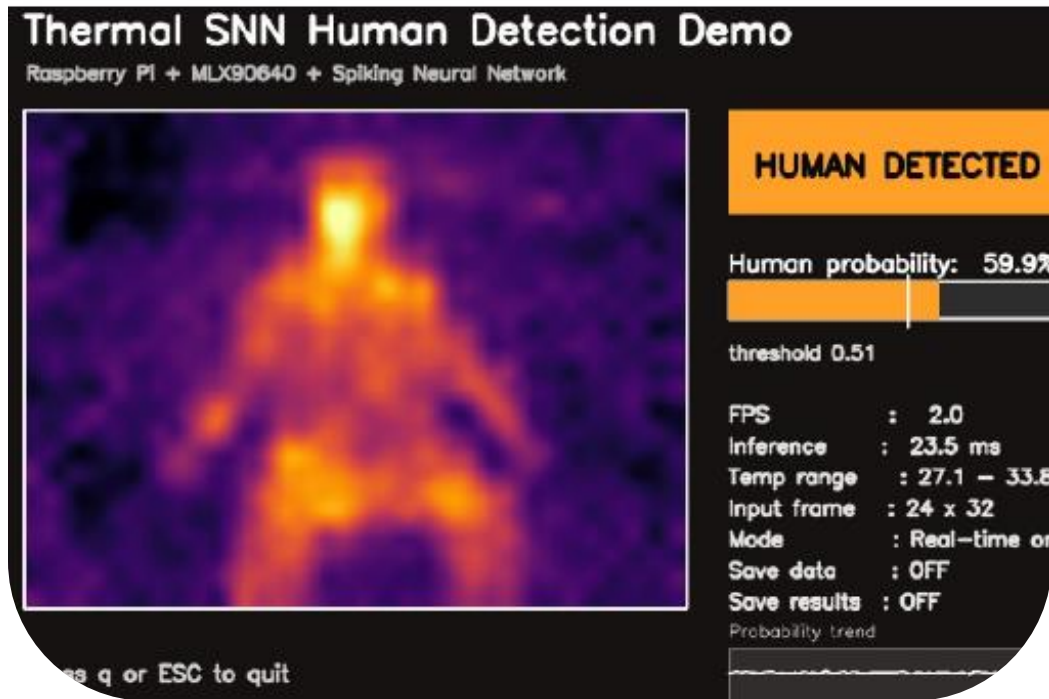
False positives

THRESHOLD 0.12: F1 = 0.9619

Recall = 0.9667 | FPR = 0.0433

Raspberry Pi SNN Deployment

Real-Time Inference GUI



PRELIMINARY DEPLOYMENT

- Capture MLX90640 frame
- Normalize on Raspberry Pi
- Run SNN inference locally
- Display probability and class

Research priority

Dynamic SNN test

| Limitations

ONE SUBJECT

Body type ▪ Clothing ▪ Behavior

ONE ROOM

Temperature ▪ Background

FRAME-LEVEL SPLIT

Similar neighboring frames

THRESHOLD SELECTION

0.12 selected without a separate
validation set

NO POWER DATA YET

No low-power claim yet

| Future Work

AUG–SEP 2026

**SNN dynamic test on
600 saved frames**

OCT–NOV 2026

**Measure CNN vs SNN
power and energy**

DEC 2026–JAN 2027

**Trade-off analysis and
thesis writing**

JAN–FEB 2027

**Thesis and
defense slides**

Thank you for your attention!